



Contents lists available at ScienceDirect

Process Safety and Environmental Protection

journal homepage: www.elsevier.com/locate/psepiChemE
ADVANCING
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Mathematical modelling and computer simulation of toxic gas building infiltration

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ARTICLE INFO

Article history:

Received 4 June 2017

Received in revised form 21 August 2017

Accepted 23 August 2017

Available online 12 September 2017

Keywords:

Consequence analysis

Infiltration

Hydrogen sulphide

Multi-zone

Modelling

CFD

ABSTRACT

A toxic gas release, e.g. H₂S, from pipeline accidents or sour wells, although improbable, may lead to serious consequences for the health of people and the environment. Such incidents might also jeopardize occupants of nearby indoor environments via infiltration of toxic contaminants. Despite that risk, there is still a lack of data and of comparative studies concerning the appropriate models and mitigation methods.

The purpose of this work is, therefore, the improved modelling of toxic gas building infiltration, by a combination of tools and approaches. Using a realistic release scenario, the present comprehensive analysis demonstrates the need to consider the detailed building characteristics and meteorology. Significant deviations are observed between simple and advanced building ingress models. Furthermore, the consequences assessment leads to contradicting conclusions depending on the employed dose-response approaches. The proposed methodology could serve as a guide for the improvement of relevant risk-assessment tools and of future studies.

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1. Introduction

Although the majority of incidents involving toxic gas release in oil and gas industry affects outdoor environments, industrial accidents such as those of Poza Rica of Mexico (24-Nov-1950, 25 died) (McCabe and Clayton, 1952), Denver city of Texas (02-Feb-1975, 9 died) (Layton et al., 1983), Lodgepole in Alberta of Canada (17-Oct-1982, 3 died) (Leahey and Schroeder, 1986) and the relatively recent in the town of Gao-qiao, Kai Xian of China (23-Dec-2003, 243 died) (Jianwen et al., 2014), support the fact that the nearby buildings and indoor environments are also at risk with possible serious consequences for the occupants. Non-process areas such as administration buildings and warehouses are often the least protected, despite being in the vicinity of potential sources. Outdoor contaminants can penetrate indoors through the ventilation system, and any kind of openings. The process of air infiltration between building and outdoor environment is known as ingress. Building ingress depends on wind pressures, ventilation system, wind turbulence and buoyancy, as well as on the leakage characteristics of

the building envelope. Other driving factors which affect the infiltration rates and indoor concentration levels are the building height and the effect of sheltering from the surrounding buildings. The latter factor can reduce the speed of the exposed wind on the building exterior, while the building height has significant effect in building ventilation, due to the buoyancy forces and wind speed to which a building is exposed (Hall and Spanton, 2012). The aforementioned factors can lead to uncertainties when trying to quantify them. Therefore, it is necessary to select a suitable method for the investigation of toxic gas building ingress.

Sour natural gas (>7 ppmv H₂S) is commonly found in many exploitations, in particular to deep and high-pressure deposits (e.g. North America, Central Asia) and occupies nearly 40% of the global gas reserves (Total, 2007). Moreover, the H₂S level for most of the natural gas wells increases along with the “age” of the reserve. Hydrogen sulphide is flammable, corrosive, with a characteristic rotten egg smell, heavier (denser) than air (specific gravity of ~1.19 at 15 °C) and extremely poisonous (fatal) even at low concentrations (e.g. 76 ppmv for a 10 min

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exposure) (<http://naturalgas.org/naturalgas/processing-ng/>). The characteristic rotten egg odour can be detected at concentrations less than 0.01 ppmv, while a loss of odour detection capability occurs at 150 ppmv or after a prolonged exposure above 20 ppmv. A dosage of more than 1000 ppmv could have acute fatal effects for the population (Leahey and Schroeder, 1986). Natural gas is mainly transported by long pipelines which pose a potential risk for the nearby communities in case of a leak or rupture. Only in Alberta, Canada, there are 18,000 km of operating sour gas pipelines and more than 6000 sour gas wells. Although, the probability of a pipeline leak or rupture is very low (making it an unlikely event), it is still possible, and then the eminent consequences, of the toxic release, can be severe and thus this risk should not be ignored.

In the unlikely event of a sour gas release combined with fire, the transport of hydrogen sulphide along with other combustion products (e.g. sulphur oxides) by a wind-blown plume can distribute poisonous materials over a large area and may lead to serious consequences for the health of people and environment (Argyropoulos et al., 2013, 2010; Assael and Kakosimos, 2010; Markatos et al., 2009). More details in relation to the health impact of H₂S exposure on oil and gas workers and surrounding population may be found in the works of Hessel et al. (1997) and Lambert et al. (2006).

In order to model the indoor exposure, there are three types of indoor air quality models: statistical regression (Valero et al., 2009), micro-environmental (Duan, 1982) and Computational Fluid Dynamics (CFD) models (Béghein et al., 2005; Choi and Edwards, 2012). The first category uses empirical techniques in order to relate, via simple analytical expressions, the indoor environment exposure to parameters such as contaminant concentration, source strength and building characteristic. Micro-environmental class of models is further classified into the measurement-based (Kornartit et al., 2010; Ozkaynak et al., 2007), mass-balance (Dimitroulopoulou et al., 2006; Gerharz et al., 2009), multi-zone (Srebric et al., 2008; Wang and Chen, 2007; Wang et al., 2010) and sub-zonal models (Megri and Haghighat, 2007; Stewart and Ren, 2006). These models assume that the pollutant concentration is well mixed within the micro-environments and their results can be used relatively easily to compare with exposure data. However, it is known that the “well mixed” assumption does not always hold true (Wang and Chen, 2008). To surpass this difficulty, advanced CFD techniques are used for modelling the indoor air flow using Large Eddy Simulation (LES) techniques (Choi and Edwards, 2008; Emmerich and McGrattan, 1998) or less computational demanding approaches such as Reynolds-Averaged Navier-Stokes (RANS) (Chen et al., 2006; Karadimou and Markatos, 2016) and Unsteady Reynolds-Averaged Navier-Stokes (URANS) (Cehlin et al., 2014). Comparison of the models can be found in the work of Deevy et al. (2008) the review papers by Milner et al. (2011) and Wang and Zhai (2016).

Early attempts to investigate the occupants' exposure to air contaminants in industrial buildings were performed by Andreopoulos et al. (1992), Papakonstantinou et al. (2000) and Ren and Stewart (2005). In the latter work, a modified version of COMIS with sub-zones (COWZ) model was used for assessing personal exposure. The numerical results were compared with available experimental and numerical (CFD) data, indicating the importance of the location and orientation of the occupant. Chan et al. (2007) proposed two metrics for quantifying the performance of shelter-in-place (SIP) as protective action for the population during an unintentional toxic release. The two introduced metrics are the casualty reduction factor and safety-factor multiplier, which adopt the nonlinear dose response effect. The proposed methodology uses a combination of models for the pollutant dispersion (Gaussian model), the indoor concentrations (well-mixed box model), and health effects (power-law toxic load model). Several other studies have also been devoted to assess the SIP effective mitigation measure during the release of a hazardous airborne material (Kulmala et al., 2016; Mannan and Kilpatrick, 2000; Sorensen et al., 2004) and to provide analytical expression for the exposure in buildings or shelters (Parker et al., 2014; Parker and Coffey, 2011) and building ventilation strategies for the protection of the occupants (Stewart-Evans, 2014).

Kassomenos et al. (2008) developed a CFD model for assessing the indoor impact to the toxic Vinyl Chloride Monomer (VCM) in a chemical plant. The model was implemented using the commercial computer program PHOENICS, as used earlier by Andreopoulos et al. (1992). Their numerical results showed that the CFD model is capable of assessing the occupational exposure. Zhang and Chen (2010) presented a combined numerical approach for quantifying the risk from an accidental toxic gas (H₂S) release. The method comprises CFD and dose-response models. The former is used for dispersion modelling, while the latter is adopted for estimating the fatality percentage of the exposed personnel. Recently, Ashraf et al. (2016) presented a combined numerical methodology for computing building ingress during a toxic gas release from a natural gas feed pipeline. The study exhibits a consequence analysis based on the predicted indoor H₂S concentration.

The above mentioned studies adopt either sub-zones or CFD models for the occupants' exposure, excluding the work of Ashraf et al. (2016) which utilises a combination of multi-zone and CFD models. However, the main focus of these works is on simple quantifying approaches for risk assessment. In the present study, advantage is taken of the manual coupling of multi-zone and CFD models, while the proposed methodology implements advanced dose-response techniques for quantifying the occupants' exposure. In addition, the proposed methodology is faster than the advanced CFD approaches and as a result it can be used as a tool to assist the preparedness and response for a hazardous material release event, as well as a guide for the improvement of relevant risk assessment tools.

The purpose of the present effort is to improve the modelling of toxic gases building infiltration, thus a specific numerical study for predicting the H₂S ingress in a non-process building of industrial area has been selected. A combination of multi-zone and CFD models, such as CONTAM (Wang et al., 2010), SLAB (Ermak, 1990; Ermak et al., 1982) and Quick Urban & Industrial Complex (QUIC) (Nelson and Brown, 2013), together with available meteorological data are employed for investigating the ingress dynamics. Furthermore, a number of dosage procedures along with various scenarios are examined for proposing adequate mitigation measures, in order to prevent any health impact on the working personnel.

2. The proposed methodology

Consequences analysis of a toxic gas incident involves estimation of the release, atmospheric transport, (potentially) building ingress, and health effects. The present study focuses mainly on last two, therefore to cover the other two use is made of available information for an H₂S release scenario and the well-established atmospheric transport model SLAB by United States Environmental Protection Agency (US EPA) (Ermak, 1990; Ermak et al., 1982). SLAB (see Section 2.1) predicts the outdoor contaminant concentration and provides information on the time taken for the plume to reach the building, for example, the maximum concentration that building is exposed to as well as the duration of the exposure. Then, the predicted values of outdoor concentration are used as input parameter for the building ingress model. For the latter, CONTAM (Wang et al., 2010; see Section 2.2) was selected, and the impact of the external wind pressures (see Section 2.3), the most critical parameter for building infiltration, was investigated by using the simple American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) correlation (Swami and Chandra, 1998) and a CFD model QUIC (Nelson and Brown, 2013). The former applies to low rise single and isolated cubical buildings, while the latter one resolves the detailed building geometry and predicts numerically the wind field around it. Finally, different mitigation scenarios and the expected health effects were assessed using the Acute Expo-

sure Guidelines by US EPA (<https://www.epa.gov/aegl>) and the probit fatality levels (Bliss, 1934; Finney, 1977).

2.1. Dispersion modelling

The estimation of the pollutant dispersion from the full bore rupture of the pipeline was modelled using the SLAB model. In the present study, the SLAB model was selected to simulate the H₂S plume and predict the concentration levels with respect to time. It is one of the US EPA alternative models and appropriate for denser-than-air gases. It was developed by Ermak (1990) and Ermak et al. (1982) and it is suitable for continuous, among others, release scenarios (Ashraf et al., 2016; Mannan, 2012). SLAB is based on the set of partial differential equations expressing the conservation of mass, momentum, chemical species and energy, in time t and downwind distance x . Details of the model equations are given in Ermak (1990) and Mannan (2012). The model has also been extensively validated with available data obtained from lab- and field-scale gas dispersion experiments, presenting satisfactory agreement (Ermak, 1990).

2.2. Ingress modelling

CONTAM (Wang and Chen, 2007; Wang et al., 2010) is a multi-zone indoor model, suitable for the determination of airflows and pressures, personal exposure under different specified conditions and contaminant concentrations. It has been widely used as indoor air quality (IAQ) model by many researchers and engineers for investigating indoor concentrations, relative humidity, PM₁₀ and PM_{2.5} in buildings (Argyropoulos et al., 2016; Fabian et al., 2012; Shrubsole et al., 2012). CONTAM also takes into account wind pressures acting on the exterior of the building as well as buoyancy effects induced by the different indoor and outdoor air temperatures. It is based on the algorithm developed in AIRNET (Walton, 1989), an airflow network simulation program developed by National Institute of Standard and Technology. The algorithm solves air flow rate from one zone to another as a function of pressure drop along a flow path (Dols and Polidoro, 2015).

Infiltration, which is the result of air flowing through flow paths or an airflow element is assumed to be governed by Bernoulli's equation (Walton and Dols, 2005). CONTAM is equipped with many available models for simulating a crack or opening in the building, ducts, fans and doorways. Among them are the power-law model, the quadratic model and the multiple opening model.

2.3. Building pressures

2.3.1. ASHRAE formulation

The assessment of the wind pressure forces acting on the external perimeter of the building affects the infiltration of the air from the outdoor to the indoor environment. Initially, the ASHRAE correlation for low-rise rectangular buildings was considered in the CONTAM model to calculate the variable wind pressure acting on the exterior perimeter of the building. The correlation calculates wind pressure coefficients averaged over a building façade as function of wind direction and is given by the following equation (Swami and Chandra, 1998):

$$C_p = C_{p0} \ln(1.248 - 0.703 \sin(a/2) - 1.175 \sin^2(a) - 0.131 \sin^3(2aG) + 0.769 \cos(a/2) + 0.07G^2 \sin^2(a/2) + 0.717 \cos^2(a/2)) \quad (1)$$

where C_{p0} ($= 0.6$) is the wind pressure coefficient, G is the natural log of the ratio of the wall length and α is the wind direction measured normal to the wall.

2.3.2. QUIC model

QUIC is a fast response urban dispersion modelling system which can compute transport and dispersion of different types of airborne pollutants within tens of seconds or minutes taking into account the effects of buildings in an approximate way. QUIC is comprised by three sub-models, namely the QUIC-URB (3-D empirical-diagnostic wind model (Röckle, 1990; Sherman, 1978)), the QUIC-PLUME (Lagrangian dispersion model) and the QUIC-CFD model. The last one consists of RANS equations which are solved explicitly in time until a steady state is reached using a projection method (Gowardhan et al., 2011). QUIC also employs a simplified zero-equation turbulence model based on the Prandtl's mixing length model (Argyropoulos and Markatos, 2015; Prandtl, 1925). During the past few years, the model has been improved and validated extensively by many scientists and engineers (Gowardhan et al., 2011; Nelson et al., 2016; Neophytou et al., 2011; Singh et al., 2008).

In the present study, it is used to calculate the values of the wind pressure coefficient in order to improve the accuracy of the obtained values, as it takes into account the building geometry while calculating the wind pressure acting on the building openings. The pressure coefficient (C_p) is defined as follows:

$$C_p = \frac{P - P_0}{0.5 \cdot \rho \cdot V_h^2} \quad (2)$$

here V is the wind speed at building height (h) (m/s), ρ is the density of air (kg/m³), P_0 is the static reference pressure (Pa) and P is the static pressure at a given location (point) on the building exterior perimeter (Pa).

The 3-D computational domain for the simulations extends to 200 m length (X), 200 m width (Y) and 20 m height (Z). A uniform Cartesian grid was also selected with 1 m resolution in all directions. In meteorological conditions, a logarithmic profile was used for the wind velocity along with 8 different wind directions from 0° to 315° with an increment of 45° and three values of wind velocity, 3 ms⁻¹, 4 ms⁻¹ and 5 ms⁻¹. A value of 0.1 m was adopted for the surface roughness, while the values of Monin–Obukhov length scale ($1/L$) (Seinfeld and Pandis, 2016) are taken equal to 0 (D-class), 0.071 (F-class) and −0.01 (C-class), respectively.

3. Application of developed methodology to a case study

3.1. Description of accident scenario and building layout

To demonstrate the proposed approach and study its sensitivity, we explored scenarios based on actual data from a natural gas plant, but the precise location and detailed description of the plant are not important, hence not disclosed herein. The characteristics of the building considered are as follows. A two-storey building with a total area of 3438.95 m² placed 1070 m away from the natural gas feed pipeline of 38 in. (0.9652 m) and 92 km length. On the ground floor there are 94 rooms including a corridor (876 m²), while on the first floor there are 103 rooms and a corridor (1605 m²). In the perimeter

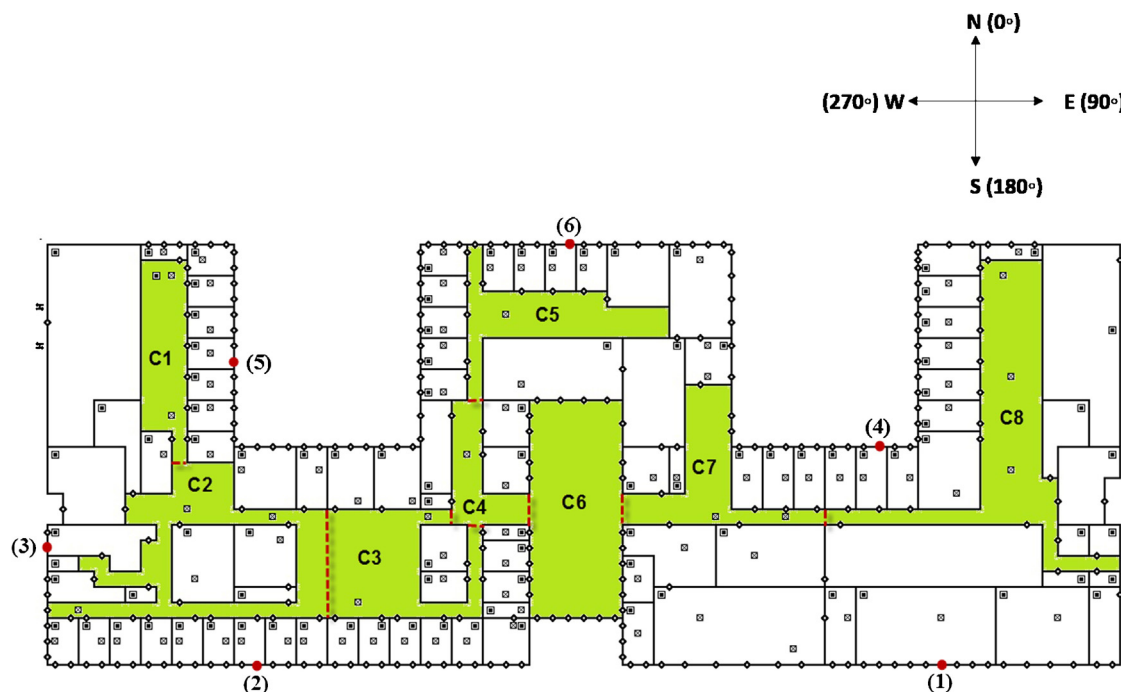


Fig. 1 – Building layout for ground floor. Numbers in parenthesis correspond to specific locations selected to be discussed in the results section. The green shared area corresponds to the full corridor configuration and C# correspond to the divided corridor Zones (with red dashed lines). (For interpretation of the references to colour in this figure legend and in the text, the reader is referred to the web version of this article.)

Table 1 – All parameters for the considered scenarios (1296 combinations).

Building envelope		Meteorology				Pressure model	
Entrance doors	HVAC	Stability class	Wind velocity	Wind direction ^b			
Closed Doors [CD]	On	C	3 m s ⁻¹	0°	135°	270°	ASHRAE
Open Doors [OD]	Off	D	4 m s ⁻¹	45°	180°	315°	QUIC-URB
	On/off ^a	F	5 m s ⁻¹	90°	225°		QUIC-CFD

^a Initially 2 min open and then closed.

^b Represents building orientation rather than the actual wind direction.

of the building, all the rooms have windows which connect with the outdoor environment. In the present study, all the windows are assumed to be closed with a leakage area of 1.73 cm²/m, while the leakage area of 187.5 cm² per door is also selected for the case of sealed scenario (Liu and Nazaroff, 2001). The heating, ventilation and air conditioning (HVAC) system has fresh air intake with a rate of 2.11 kg s⁻¹ (~82% recirculation with ~3.4 air exchange rates per hour (ACH)) and the intake height is taken at 7.9 m from the ground.

Fig. 1 depicts the building layout for the ground floor. This building layout has been produced by CONTAM model according to the available HVAC data. Similar approach is also adopted for the building layout reconstruction of the first floor. Table 1 presents the selected parameters for the parametric analysis, each combination reflects a studied scenario of 1296 total.

The corridor of the building will prove to be the most vulnerable area; therefore we included two different configurations (Fig. 1), a full (continuous) corridor and one divided into eight smaller zones (red dashed lines).

We assume a release of high pressure sour gas, of around 0.77% (w/w) H₂S in methane, due to full bore rupture. Leak duration of 60 min and a mass source rate equal to 11,500 kg/s are considered. The release of H₂S is directed to the non-process building, while the temperature and pressure of

pipeline are assumed to be maintained at 27 °C and 83 bars, respectively. In this study, the reactivity of the H₂S with indoor surfaces, building ingress pathways and air-handling filters is not considered. However, the interested reader is directed to the work of Liu and Nazaroff (2001) for reactive gas penetration.

Finally, the range of the selected meteorological parameters (see Table 1) are based on a previous study on the modelling of dense gas dispersion in Qatar (Gopalaswami et al., 2015).

3.2. Mitigation scenarios

Only “evacuation” was considered as the mitigation scenario in the present study. Two simple mitigation – evacuation – scenarios were considered. The first scenario investigates the case of escaping from the building (running velocity ~1.4 m s⁻¹) in the x- or y-direction after the release event. The second scenario assumes that the personnel remain indoors during the entire period of the event. In either case, a dose-response approach is necessary to estimate the consequences. Herein, the probit analysis and toxic load algorithm were employed.

The probit analysis (Bliss, 1934; Finney, 1977) is based on acute inhalation toxicity data and estimates the percentage

fatality during a toxic release event. The equations for this method can be written as follows (Ten Berge et al., 1986):

$$Y = b_0 + b_2 \ln k \quad (3)$$

$$P = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{Y-5} \exp\left(-\frac{1}{2}u^2\right) du \quad (4)$$

where k is given by the following equation:

$$k = C^n t \quad (5)$$

where Y is the probit function, P is the fatality frequency, k is the toxic dose, C is the concentration, t is the duration and $n = b_1/b_2$. The b_0 ($= -10.87$), b_2 ($= 1$) and n ($= 1.9$) are empirical parameters based on the considered gas (e.g. H_2S) and their values are adopted from TNO probit function (HSE, 2013).

The toxic load algorithm (Boris and Patnaik, 2014; Oran and Boris, 2001; Stage, 2004) is based on the Acute Exposure Guideline Levels (AEGLs, (NRC, 2013)) and the quantification of the toxic dosage using the non-linear equation of Ten Berge et al. (1986). The AEGL determine the dosages, i.e. constant concentrations for given durations of 10 min to 8 h, presenting three levels of harm or the general public, namely notable discomfort, disability and life threatening or death (HSE, 2013; NRC, 2013).

The numerical approach accounts for the time slots for AEGL-1 (i.e. 0.51 ppm for 60 min exposure), AEGL-2 (i.e. 27 ppm for 60 min exposure) and AEGL-3 (i.e. 50 ppm for 60 min exposure) and defines a power law profile with reference to the rate of accumulation of toxic load (Boris and Patnaik, 2014). The approach can be determined by the following expressions:

$$TL(t) = \int_0^t TL_{rate}(t') dt' \quad (6)$$

$$TL_{rate}(t) \equiv \frac{dTL}{dt} = \frac{1}{t_b} \left[\frac{C(t)}{C_{t_b}} \right]^n \quad (7)$$

here $TL(t)$ is the integrated toxic load, t is the time, t_b is the reference AEGL time band exposure step, C is the concentration and n ($= -0.23$ for H_2S) is the power exponent.

4. Results and discussions

4.1. Plume rise and pollutant dispersion

In this study, the plume height and pollutant concentration were calculated for nine different scenarios (Table 1). For the sake of clarity, we use the “F3”, “D5” and “C4” terms for the atmospheric stability class and wind velocity (i.e. [Stability Class] [wind velocity]).

Fig. 2 illustrates the maximum centreline concentration (MCC) of the plume along with the ground level concentration (GLC; at 2 m height) downwind from the source. The Relative Humidity (RH) parameter has almost negligible effect on the maximum concentration and the plume height; therefore we adopted a value of 50%. Also, “F3” appears to lead to the highest MCC values of H_2S and “C4” the lowest. In the vicinity of the source, the range of MCC for the three types of stability class is between 9500 ppmv and 9700 ppmv, approximately. Moreover, the values of MCC gradually decrease to less than 900 ppmv as

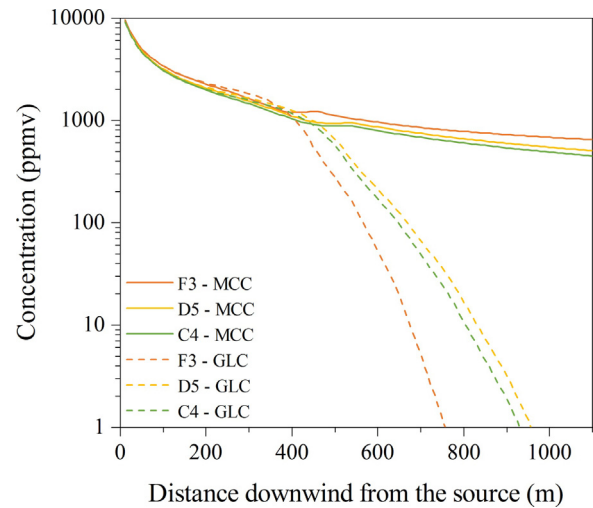


Fig. 2 – Maximum centreline concentration and ground level concentration for RH = 50% and “F3”, “D5”, and “C4” (i.e. [Stability Class] [wind velocity]).

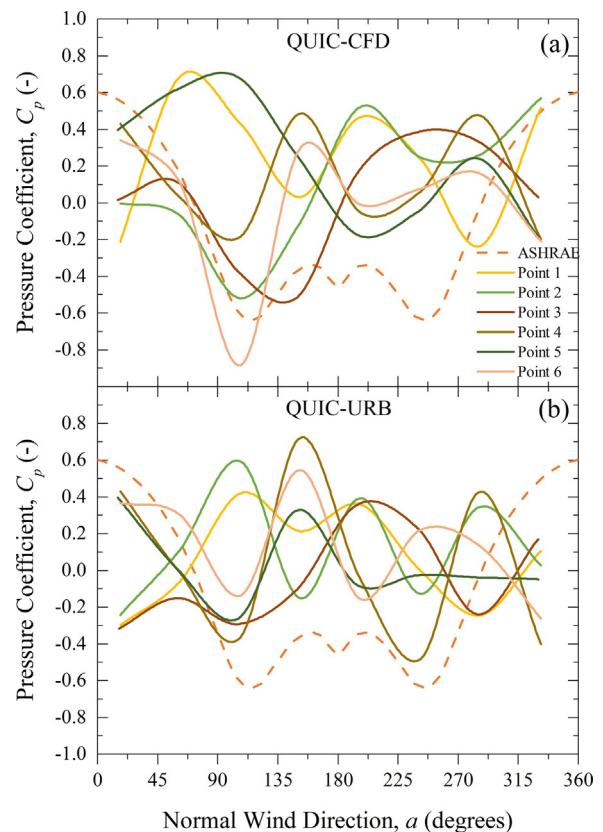


Fig. 3 – (a) C_p vs. normal wind direction calculated by QUIC-CFD and (b) C_p vs. normal wind direction calculated by QUIC-URB.

the distance from the source release increases up to 1200 m, approximately.

In the case of GLC, it is seen that close to the release source the values vary from 9500 ppmv to 9700 ppmv, similar to the case for MCC. However, the values of GLC significantly decrease to less than 1200 ppmv as the distance from the emission source reaches to 400 m. It is also noted that there is a significant difference in concentration levels between the MCC and GLC near the building. This behaviour can be explained by the rise of the sour gas buoyant plume. Just for the sake of consequences analysis, we use the higher values i.e. the MCC.

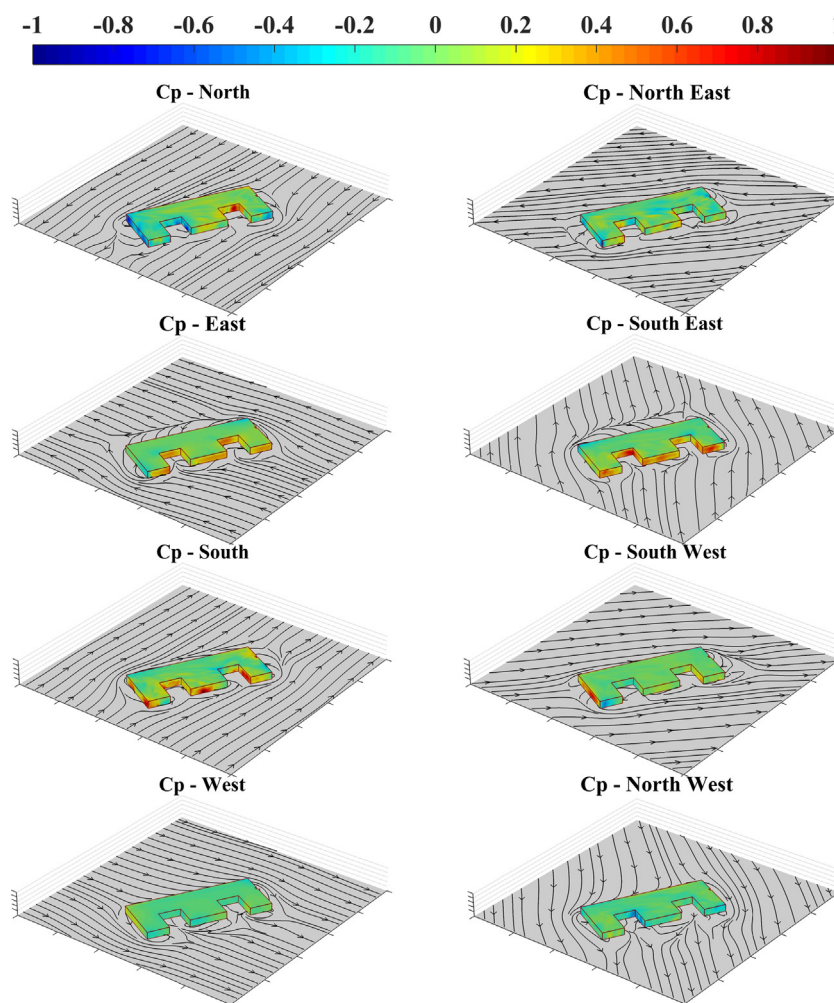


Fig. 4 – Surface streamlines and C_p for “D5” (i.e. [Stability Class] [wind velocity]) and 8 different wind directions calculated by QUIC-CFD.

4.2. Calculation of building pressure field

Fig. 3 presents the variation of the wind pressure coefficient, C_p , on a surface against its orientation (e.g. 0° correspond to a wind direction normal (perpendicular) to the surface). C_p , as per ASHRAE formulation (dashed line) receives its highest values at normal wind direction and its lowest ones at parallel winds. Note also the building wake effect on C_p for the range of 135° – 225° which values are still negative but higher than the parallel wind case. However, ASHRAE formulation corresponds to a single and isolated cubical building and so we observe significant differences when we calculate C_p using any of the models that resolve the exact building geometry. As an example, we selected six representative building locations (see Fig. 1 points 1–6) and we plot C_p for QUIC-CFD (Fig. 3a) and QUIC-URB (Fig. 3b) versus the orientation of the surface. The differences observed are attributed to the complicated wind field around the building. As expected, the global wind direction varies from the local wind direction and thus the effect on C_p is not the same as per the ASHRAE formulation. Fig. 4 depicts the C_p on the building envelope for eight different wind directions, at 5 ms^{-1} wind speed and under neutral atmospheric stability. Following up on the observed C_p variations, we expect that any addition of neighbouring buildings or obstacles, as well as the use of a more advanced fluid dynamics tool, they will also affect the C_p values.

4.3. Prediction of indoor concentrations

Fig. 5 illustrates the concentrations of the two corridor configurations (full and divided). It is observed (Fig. 5a–b) that the values of concentration are significantly higher for Zone C7 (Fig. 1) than for the full corridor configuration for each calculation method, while QUIC-CFD predicts the highest values. The stability class and wind velocity also affect the concentration for both corridor configurations. In the full corridor configuration, the “F3” appears smaller concentration values than the “D5”, while for Zone C7 is the opposite, excluding the case for ASHRAE model. In the case of Zone C7, it is seen that the values of concentration of “D5” are lower than for “F3” using QUIC-URB and QUIC-CFD models. This occurs due to the influence of Zone C7 location (Fig. 1) in the building and the specific wind direction which does not affect at the same way as in the case of ASHRAE model. The deviation of indoor concentrations for the two different corridor configurations supports the fact that the well-mixed assumption in CONTAM introduces an error for large areas.

Fig. 5c–d illustrate the concentrations for the scenario of CD-HVAC on. It is observed that the difference between the concentration values obtained from the different models at the same time and the two corridor configurations are small. However, the concentration values for “F3” are higher than “D5”. This is in accordance with the outdoor concentrations

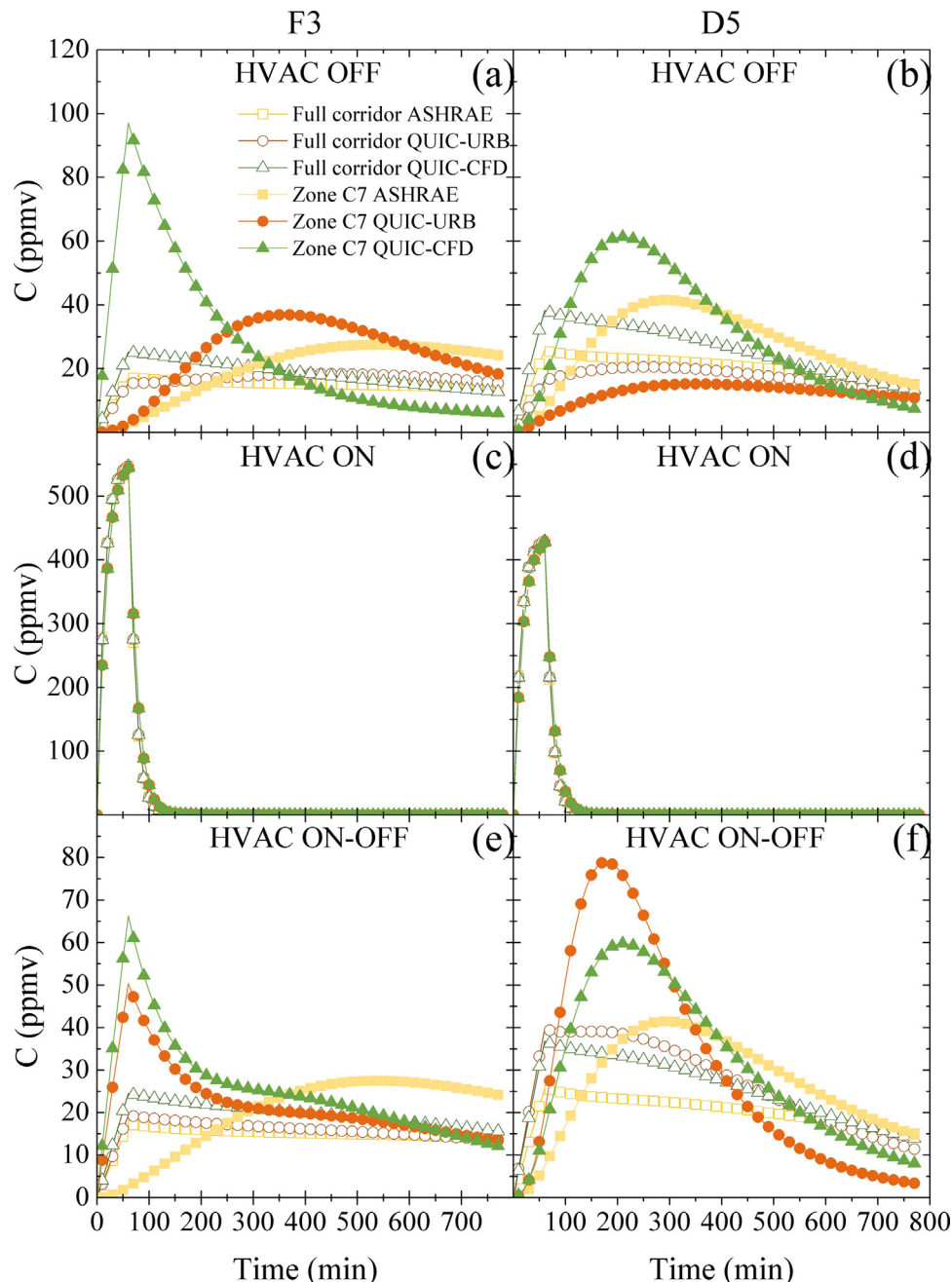


Fig. 5 – Indoor concentrations of the two corridor configurations at 180° with “F3” and “D5” (i.e. [Stability Class] [wind velocity]): (a)–(b) scenario of CD-HVAC off, (c)–(d) scenario of CD-HVAC on, and (e)–(f) scenario of CD-HVAC on/off.

and it was expected, since the HVAC operation is on and allows the entrance of the outdoor contaminant in the building.

For the scenario of CD-HVAC on/off (Fig. 5e–f), the highest value of concentration appears in Zone C7 with QUIC-CFD model for “F3” and the Zone C7 with QUIC-URB model for “D5”. The concentration values for “D5” are higher than for “F3” at the most of the time excluding the case of Zone C7 with QUIC-CFD model. It is also noticed that the corridor configuration appears to lead to higher values for the divided (Zone C7) compared to the full.

The wind direction is also an important factor of the ingress phenomenon, as the dispersion of contaminant from outdoor to indoor follows a specific wind direction. Fig. 6a presents the concentration for the full corridor configuration using the ASHRAE model, while the highest value of concentration is observed for the 180° at 12 ppmv. On the contrary, Fig. 6b illustrates that the Zone C7 appears to lead to higher range of

values for the most of the wind directions which supports again the invalidity of “well mixed” assumption between the full and divided corridor configuration.

In Fig. 6d, it is shown that the QUIC-URB model displays the highest value of concentration for Zone C7 at 60 ppmv and 90°. Similar behaviour can be observed from Fig. 6b. This occurs because the direction of wind is normal to the entrance doors of the building which is directly connected to the corridor area. The HVAC air inlets which supply the outdoor air for the first 2 min are also located normal to the 90° and as a result an increased amount of gas enters the building.

Fig. 6e–f depict the predictions for the QUIC-CFD model which accounts for the wind pressures on the facade of the building, presenting a more realistic representation of the ingress phenomenon. It is noticed that the highest value of contaminant concentration is calculated for Zone C7 at 60 ppmv and 315°. The concentration values are also higher

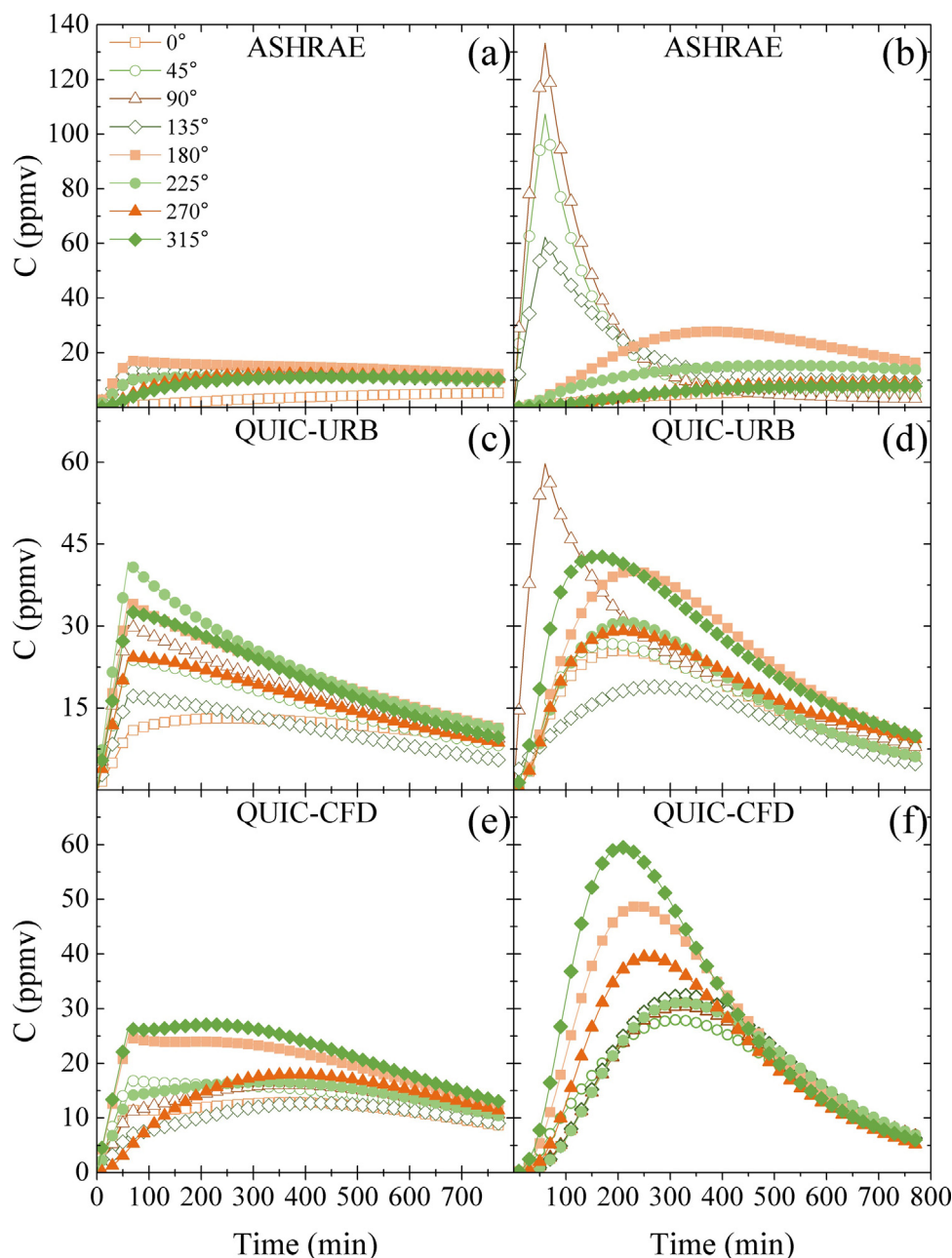


Fig. 6 – Indoor concentrations for the scenario of CD-HVAC on/off, with eight different wind angles and “C4” (i.e. [Stability Class] [wind velocity]): (a)–(b) ASHRAE, (c)–(d) QUIC-URB and (e)–(f) QUIC-CFD.

for the divided corridor compared to the full corridor configuration. Finally, it is seen that the highest concentration value is given for different wind direction compared to the previous obtained results from the ASHRAE correlation and the QUIC-URB model (Fig. 6b and d), indicating the differences between the three selected models.

4.4. Analysis of mitigation methods

Fig. 7a presents the predicted outdoor concentrations for the “F3”, “D5” and “C4” scenarios by SLAB model which have been used for quantifying the toxic effects of H_2S with the probit and toxic load methods. According to Fig. 7b, it is observed that the probit method for “F3” predicts a value of 3% fatality in x-direction 5 min from the start of the release event, while displays 0% fatality in the y-direction. It is also seen that the fatality values for cases “D5” and “C4” are approximately 1.25% and 0.5% in x-direction and 0% for all cases in y-direction.

On the contrary, the TL method leads to a value of one for all cases at less than 0.5 min (Fig. 7c). The normalised results of TL to a value of more than one indicate that the general population could experience life-threatening health effects or lethal effects. As a result, the building’s occupants are exposed to fatal conditions in a very short period of time regardless the considered meteorological conditions which make the evacuation scenario infeasible. The underestimation of the probit method can be explained by the lack of taking into account the short term exposures and the selection of the appropriate value of the parameter n (HSE, 2013) (see Eq. (5)).

To surpass this problem, we introduce an “adjusted” probit function. The adjusted probit is fitted to the H_2S AEGL-3 threshold and thus we estimate a new parameter n for the power law profile, taken equal to 2.77 (Fig. 8). In Fig. 7d, it is shown that the % adjusted probit appears 100% fatality after the first half minute of the release which apparently matches the results of the TL approach. The proposed approach to

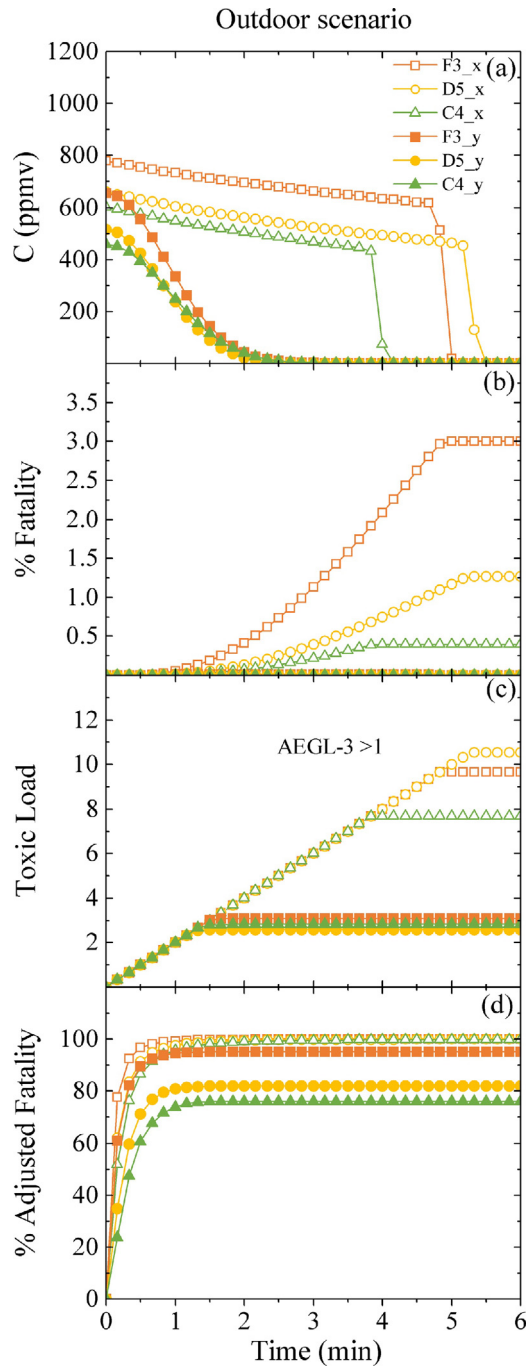


Fig. 7 – Consequences analysis for the outdoor scenario for “F3”, “D5” and “C4”: (a) concentration outdoor levels, (b) % fatality based on the Probit method, (c) fatalities based on the Toxic Load method, (d) % fatality based on the Adjusted Probit.

obtain an adjusted probit function does not replace the typical probits neither the TL approach. It offers the possibility to assess risk for the general public and working personnel with the same approach, but different parameters.

The second evacuation scenario examines the case that the personnel remains indoors. For this scenario, we estimate a “weighted probit” which expresses the total probability of fatality, defined as:

$$Y = \sum_i Y_i \cdot X_i \quad (8)$$

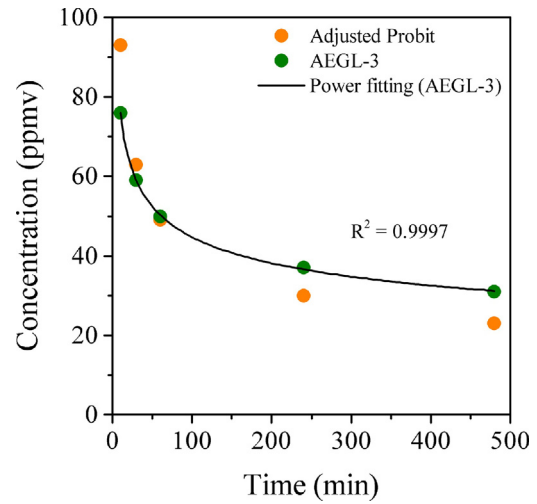


Fig. 8 – Introduction of adjusted probit function.

where Y_i is the probability of fatality at i Zone, X_i is the probability of being at a particular location i and defined as:

$$X_i = \frac{V_i}{\sum V_i} \quad (9)$$

where V_i is the volume of i Zone in the building.

Fig. 9a depicts the corridor concentrations at the ground level floor of the building for OD scenarios, while Fig. 9b presents for CD scenarios. It is observed that the concentration levels for OD scenarios with or without HVAC operation are much higher than that for CD scenarios regardless the selected model. In Fig. 9b, it is also seen that the QUIC-URB and QUIC-CFD yield to higher values of concentration than for ASHRAE model. From Fig. 9c and d, it is shown that the OD scenarios display higher % fatality than for the CD scenarios. The range of values observed is between 33% and 38% for OD cases, while for OD cases is less than 0.35%

All the OD scenarios exhibit a TL value more than one at less than 0.5 min which is fatal for the personnel (Fig. 9e). However, this is not in accordance with the CD scenarios, excluding only the scenario of CD-HVAC on/off with QUIC-URB model (Fig. 9f) which overpasses the AEGL-3 threshold after 200 min. This behaviour is also against the prediction of the % fatality based on the weighted probit method and supports the difference between the selected models for the predicted response. It is important to mention that even though the remaining CD scenarios do not exceed the AEGL-3 threshold (fatal effects) the concentration levels are still over the AEGL-2 threshold. As a result the working personnel could experience irreversible health effects or an impaired ability to escape (Stage, 2004).

In Fig. 9g, it is noticed that the adjusted probit appears to lead to 100% fatalities at less than 20 min, complementing the TL predictions (Fig. 9e). On the other hand, Fig. 9h exhibits values lower than 100% fatality for all the CD scenarios, even though we were expecting to follow again the TL predictions (100% fatality after 200 min) for the scenario of CD-HVAC on/off with QUIC-URB model.

The large number of rooms (197) of the selected realistic building makes it difficult to present results for each one of them and for all studied scenarios (i.e. wind directions). At the same time, usage of the highest (worst case) or the average (weighted) values smoothes the extent of the consequences. Therefore, we use a polar frequency plot to depict the consequences extent (distribution of AEGL) for all rooms at dif-

Indoor scenario

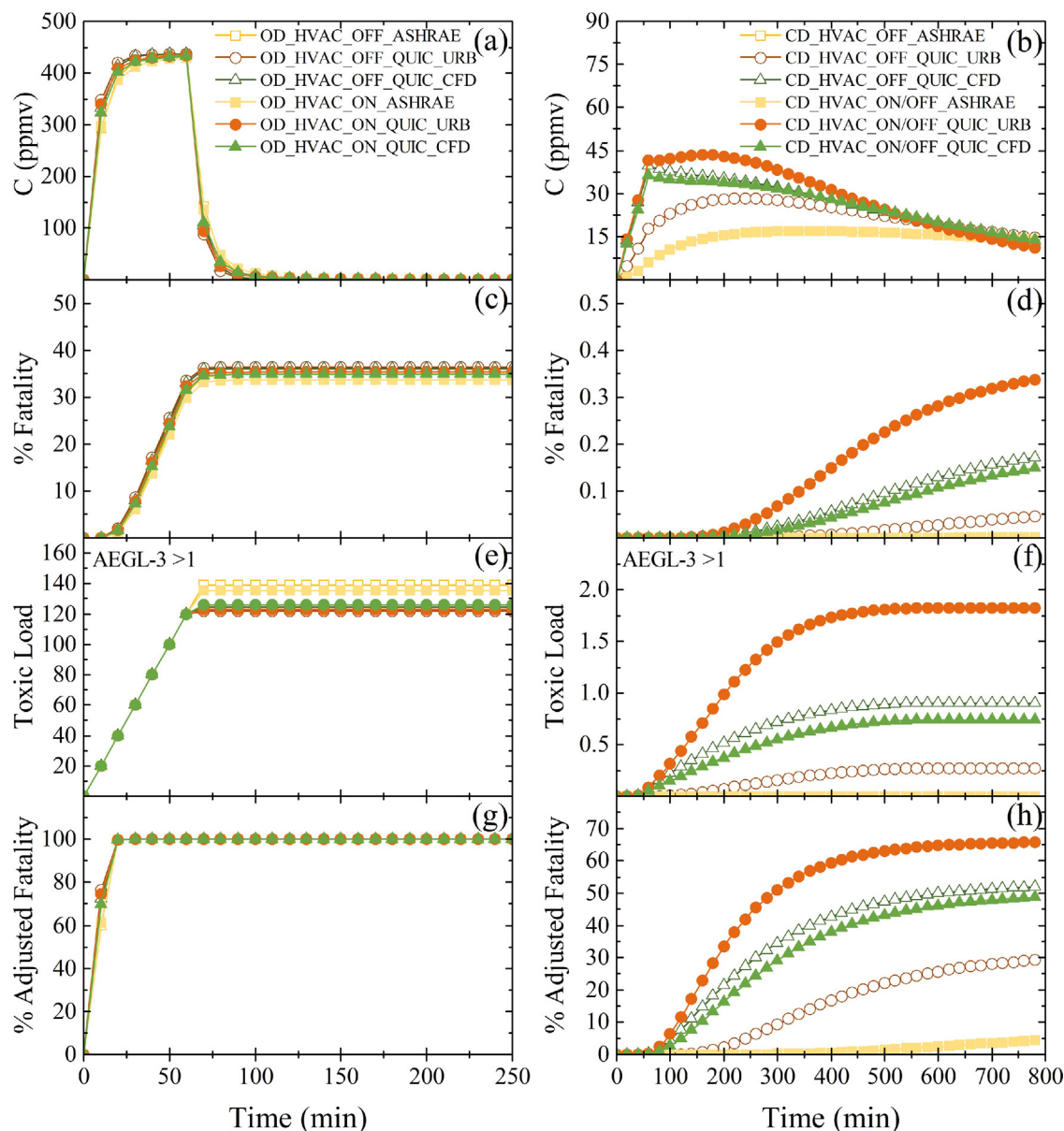


Fig. 9 – Consequences analysis for the indoor evacuation scenario for “D5” and 315°; Opened door (OD) case: (a) corridor concentration, (c) % fatality based on the weighted probit method, (e) fatalities based on the Toxic Load method, and (g) % fatality based on the Adjusted Weighted Probit. Closed door (CD) case: (b) corridor concentration, (d) % fatality based on the weighted probit method, (f) fatalities based on the Toxic Load method, and (h) % fatality based on the Adjusted Weighted Probit.

ferent wind directions. For example, Fig. 10 depicts the AEGL-3 distribution for the CD-HVAC off scenario where the radius corresponds to the value itself, the colour represents the number of rooms with a specific AEGL value at given wind direction. In Fig. 10a, it is observed that more than 60 rooms maintain a value between 0 and 0.2 (grey colour, at the centre) at any wind direction, while at 0° we observe that ~4 rooms exceed the AEGL-3 threshold (>1.0 fatal effect) and few others to have values between 0.4 and 0.9. Note that AEGL-3 distributions vary significantly for the three different models (ASHRAE, QUIC-CFD, and QUIC-URB), indicating different number of rooms with more or less risk for the working personnel. The largest number of rooms which exceeds the AEGL-3 limit occurs at 90° for QUIC-CFD, at 0° and 315° for ASHRAE, and at 90°, 135°, 270° and 315° for QUIC-URB.

Fig. 11 presents the distribution of AEGL-2 values, which limit corresponds to irreversible health effects. Again, it is seen that each model exhibits different levels of risk for a number of rooms, while for large number of rooms the AEGL-2 threshold is exceeded at all wind directions regardless the selected model. In general, QUIC-URB model predicts larger number of AEGL-2 and AEGL-3 threshold overshoots, while the ASHRAE model smaller.

5. Conclusions

A general methodology was developed for predicting toxic gas infiltration in a non-process building of industrial facilities. This methodology was applied to the case of a sour gas release. Sensitivity analysis was conducted for the most

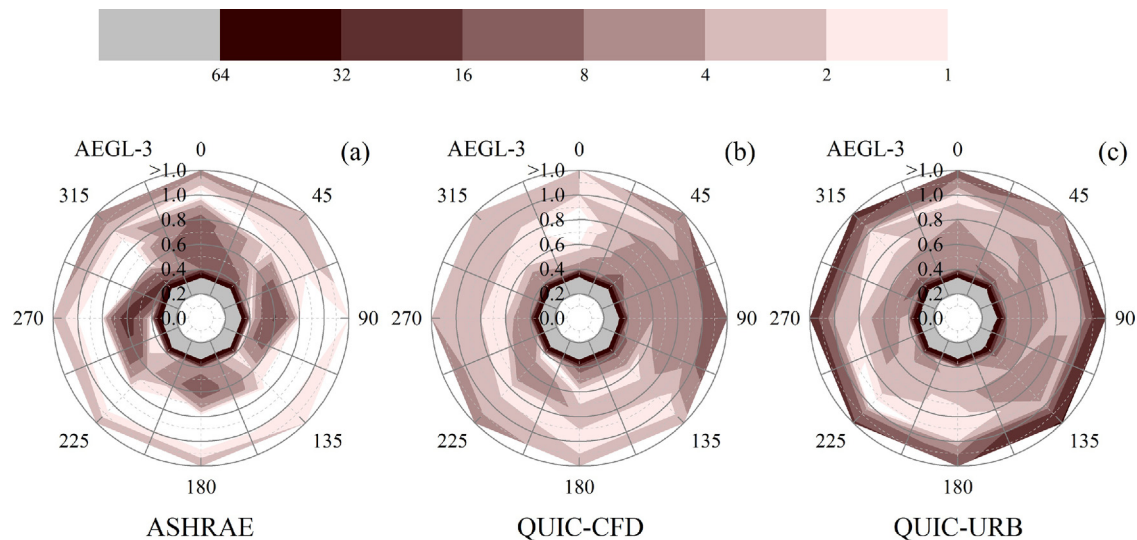


Fig. 10 – Distribution of the AEGL-3 values, for all building rooms, at different wind directions and for the CD-HVAC off scenario. Radius corresponds to the AEGL value and the colour represents the number of rooms with this value for: (a) ASHRAE model, (b) QUIC-CFD model and (c) QUIC-URB model.

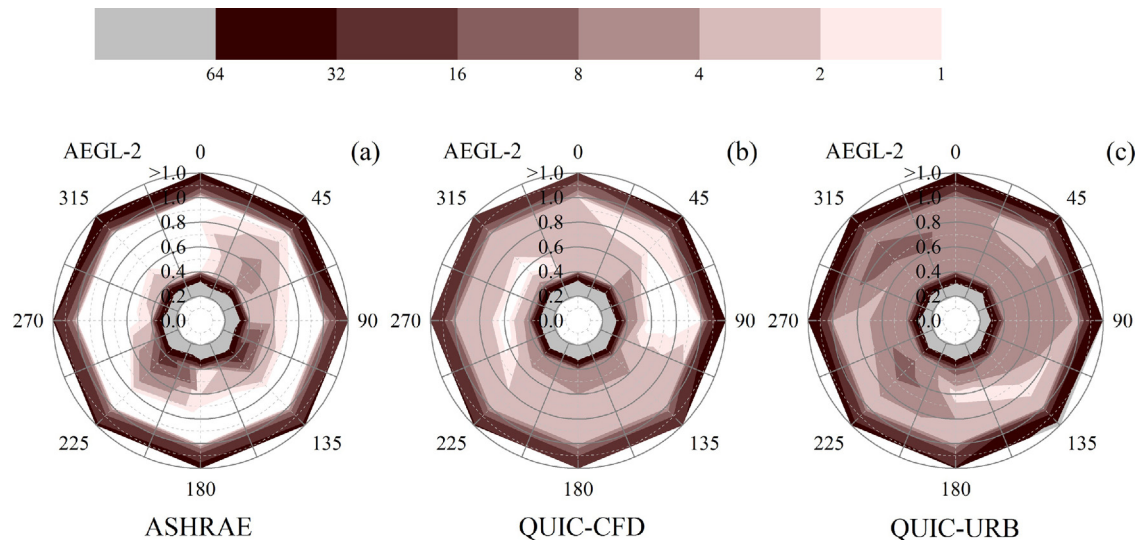


Fig. 11 – Distribution of the AEGL-2 values, for all building rooms, at different wind directions and for the CD-HVAC off scenario. Radius corresponds to the AEGL value and the colour represents the number of rooms with this value for: (a) ASHRAE model, (b) QUIC-CFD model and (c) QUIC-URB model.

important parameters: wind pressure profiles, meteorological conditions, and characteristics of the building envelope. Although a large number of different scenarios (more than 1296) were examined, the analysis is still not exhaustive and several assumptions had to be made, for example for the assumed release and mitigation scenarios.

It was found that the outdoor concentration levels exceed in all cases the AEGLs around the selected non-process building, even if far away from the source (~ 1 km). The highest values are observed for the most stable atmosphere studied ("F3" scenario), due to the limited mixing.

The results obtained at six different locations of the building indicate that there is dissimilarity in the wind-direction dependency due to the complex building geometry. In any case, the predicted pressure coefficients are within the range of ASHRAE recommendations. Given the findings of this work and the absence of experimental data, the preferred model

for the wind pressure field is the QUIC-CFD, which although significantly slower than QUIC-URB, it does give more realistic predictions for the wind pressure field and is the only model compared to the other two that accounts for turbulence effects (Gowardhan et al., 2011). On the other hand, ASHRAE correlation, developed and validated for simple cubical buildings, cannot predict accurately the wind profile for the present case due to the complex building geometry.

It is also shown that the scenarios with closed doors (CD) and HVAC off or on/off have the highest survival rate for the building occupants, while the open doors (OD) scenarios the lowest. "Shelter-in-place" (SIP) as an emergency response measure was considered through the scenarios with CD and HVAC off or on/off. However, extra attention is needed, because a number of rooms maintain high concentration of H_2S which could lead to adverse health effects. It is also important to mention that an estimation for taping and sealing a

room in order to implement SIP can hold from 3 min to 39 min, with an average of 17 min (Sorensen et al., 2004). Therefore, it is necessary to determine the amount of physical protection (Mannan and Kilpatrick, 2000) and the role of the available ventilation system (Stewart-Evans, 2014), which can be provided by a specific building before deciding SIP.

Moreover, it is demonstrated that the dose-response approach is probably the most important aspect of the mitigation analysis. The probit analysis shows negligible fatality rates inside the building, while in most cases the toxic load approach (based on the AEGL) exceeds the fatality levels. The latter is recommended for the general public and it better fits the character of the occupants of a non-process building. For this reason, we propose the use of an adjusted probit function, which reflects better the AEGL health effects but still maintains the sensitivity of the probit function.

Finally, the estimated health effects proved also sensitive to the wind pressure calculation approach. For example, the QUIC-URB model predicts higher impact than the other approaches. This variation became more profound when we employed the AEGL-2 values into the toxic load approach to estimate the irreversible or other serious, long-lasting adverse health effects or an impaired ability to escape. Therefore, it is important to select the appropriate model for each case in order to assess adequately the risk. Another good example of such a discrepancy is the ingress model itself for which we observed differences when the corridor was modelled as a full or a divided area. Such differences between models and approaches raise the need for more experimental studies, in order to validate the obtained numerical results, and for more advanced numerical methods, for the prediction of indoor toxic levels in large rooms. However, the present comparison of multiple numerical tools and the extensive amount of numerical data obtained and compared indicate that the proposed methodology is sound and, it is hoped, it may benefit the risk-assessment decision-making community.

Acknowledgements

This publication was made possible by a NPRP award [NPRP 7-674-2-252] from the Qatar National Research Fund (a member of The Qatar Foundation) and the support of the Mary Kay 'O Connor Process Safety Center at Qatar. The statements made herein are solely the responsibility of the authors. The High Performance Computing resources and services used in this work were provided by the IT Research Computing group in Texas A&M University at Qatar.

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